

NORTH CENTRAL STATE, RI, USA

WMO: 725054

Lat: 41.921N

Lon: 71.491W

Elev: 134

StdP: 99.72

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64710

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.1	(c) -12.5	(d) -24.0	(e) 0.4	(f) -13.3	(g) -21.5	(h) 0.6	(i) -10.3	(j) 10.9	(k) -4.8	(l) 9.7	(m) -3.9	(n) 3.7	(o) 300	(p) 0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 31.2	(d) 23.2	(e) 29.1	(f) 22.0	(g) 27.8	(h) 21.0	(i) 24.3	(j) 29.1	(k) 23.4	(l) 27.5	(m) 22.5	(n) 26.3	(o) 3.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	17.8	26.8	22.2	17.2	26.1	21.2	16.2	25.1	73.9	29.2	70.1	27.3	66.8	26.2	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.8	(b) 7.8	(c) 6.9	(e) -17.4	(f) 33.7	(g) 2.3	(h) 1.8	(i) -19.1	(j) 35.0	(k) -20.4	(l) 36.0	(m) -21.7	(n) 37.0	(o) -23.4	(p) 38.2		
(a) 8.8	(b) 7.8	(c) 6.9	(e) -18.7	(f) 25.9	(g) 2.2	(h) 0.8	(i) -20.3	(j) 26.5	(k) -21.6	(l) 27.0	(m) -22.8	(n) 27.5	(o) -24.4	(p) 28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.2	-1.3	2.6	8.7	13.7	18.6	21.8	21.0	17.0	11.0	5.4	0.4
	DBStd	9.49	5.91	5.43	5.22	4.37	4.10	3.68	2.87	3.03	3.68	4.40	4.81	4.93
	HDD10.0	1508	381	316	237	73	9	0	0	0	1	39	152	301
	HDD18.3	3416	637	548	487	289	156	42	4	9	68	229	387	557
	CDD10.0	1433	2	1	8	36	123	258	367	342	210	72	14	2
	CDD18.3	301	0	0	1	2	12	50	113	93	28	3	0	0
	CDH23.3	2288	0	0	4	27	137	401	871	659	173	16	0	0
	CDH26.7	603	0	0	1	8	41	103	246	169	35	1	0	0
Wind	WSAvg	2.9	3.5	3.5	3.6	3.5	3.0	2.4	2.2	2.2	2.4	2.7	3.2	3.3
Precipitation	PrecAvg	1240	103	89	127	112	94	93	90	96	105	111	102	116
	PrecMax	1466	180	198	377	205	199	291	231	165	237	358	178	206
	PrecMin	979	41	49	4	34	30	5	34	23	36	12	16	35
	PrecStd	151	35	37	76	46	40	76	43	43	52	78	39	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	15.8	22.4	27.7	31.0	32.3	32.8	32.4	30.8	25.9	19.7	16.3
		MCWB	12.7	11.4	13.3	17.2	19.1	23.7	24.5	24.9	23.2	18.0	15.2	12.7
	2%	DB	12.4	12.2	16.9	22.4	27.3	29.1	31.2	30.2	27.3	22.5	17.4	12.9
		MCWB	10.0	9.0	10.3	13.1	17.9	21.7	23.6	23.3	20.5	17.5	14.0	11.1
	5%	DB	8.9	8.9	12.8	18.9	23.8	27.6	29.0	28.1	25.2	20.5	15.4	11.1
		MCWB	6.6	6.0	8.3	11.8	16.2	20.6	22.4	21.9	19.5	16.6	12.0	9.0
	10%	DB	6.0	7.0	10.2	16.4	21.3	26.0	27.7	27.2	23.0	18.0	13.0	7.8
		MCWB	3.3	3.7	6.4	10.0	14.6	19.3	21.4	21.0	18.6	14.2	9.9	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	12.7	14.6	17.8	21.1	24.6	25.7	25.7	23.6	21.0	17.0	14.3
		MCDB	15.2	14.1	20.5	22.8	27.3	30.5	31.6	31.0	27.7	23.2	18.0	15.9
	2%	WB	10.4	9.1	11.7	15.3	19.4	23.1	24.4	24.2	22.2	19.0	14.9	11.8
		MCDB	12.1	11.5	14.9	20.5	25.3	28.1	29.3	28.4	25.5	21.1	16.9	12.9
	5%	WB	6.7	6.4	9.0	13.2	17.5	21.7	23.4	23.3	20.9	17.2	12.9	8.7
		MCDB	8.7	8.8	12.4	17.3	21.8	26.1	27.8	26.6	23.7	19.3	14.7	10.4
	10%	WB	3.6	3.6	6.7	11.4	15.6	20.4	22.5	22.3	19.7	15.2	10.7	5.6
		MCDB	5.6	6.1	10.0	15.5	20.0	24.1	26.4	25.6	22.7	18.1	13.1	7.8
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.1	9.4	10.7	10.8	10.5	10.3	10.2	10.4	10.3	8.9	8.1
		MCDBR	11.4	11.6	13.0	15.1	14.7	12.7	11.6	11.1	11.8	11.8	11.0	11.2
		MCWBR	10.0	9.2	8.6	8.4	7.6	7.1	5.7	5.6	6.8	7.7	8.7	10.4
	5% WB	MCDBR	11.3	11.2	12.3	12.5	13.1	11.2	9.9	9.6	9.8	10.2	9.9	10.8
		MCWBR	10.5	10.0	9.4	8.5	7.6	6.9	5.6	5.4	6.5	8.1	9.0	10.8
Clear-Sky Solar Irradiance	taub	0.290	0.299	0.324	0.365	0.405	0.448	0.469	0.431	0.366	0.330	0.305	0.291	
	taud	2.494	2.482	2.473	2.387	2.307	2.220	2.169	2.262	2.421	2.538	2.569	2.563	
	Ebn at Noon	868	914	926	904	872	832	809	830	867	867	846	840	
	Edh at Noon	73	86	98	115	128	140	146	129	102	81	68	63	
All-Sky Solar Radiation	RadAvg	1.86	2.73	3.73	4.78	5.27	5.75	5.98	5.37	4.36	2.91	2.00	1.54	
	RadStd	0.13	0.24	0.26	0.51	0.56	0.57	0.33	0.28	0.32	0.27	0.15	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+45

Nomenclature: See separate page